
VALIDATION REPORT

GUJARAT FLUOROCHEMICALS LTD

VALIDATION OF CDM PROJECT
FOR GHG EMISSION REDUCTION BY THERMAL
OXIDATION OF HFC23

Project No. CDM.Val0018

Date: August 12, 2004 revised 8th March 2005



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Approved by:	Organisational unit: SGS Climate Change Programme
Client: Gujarat Fluorochemicals Ltd	Client ref.:

Summary:

This validation concerns a project to be implemented by Gujarat Fluorochemicals Ltd (GFL) in India to eliminate or substantially reduce emissions of the potent greenhouse gas HFC23. The objectives of the validation exercise are to confirm that the project meets the necessary criteria, that the project follows the approved methodology and that the proposals presented by GFL will lead to a realistic determination of the quantities of HFC23 destroyed. The scope of the validation covers the baseline methodology and a study of the monitoring methodology and plan proposed for the assessment of the quantities of HFC23 destroyed and the emissions of greenhouse gases associated with the destruction process.

The validation, undertaken by SGS, involved a desk study of the Project Design Document (PDD) and the Approved Methodology, followed by a visit to the GFL site in India where key GFL personnel, advisors and consultants involved in the project were interviewed. Conformance to local legal and environmental regulations was also confirmed.

Clarifications on a number of issues were requested by SGS; these were provided satisfactorily by GFL and resulted in a revision of the original PDD.

In the opinion of SGS, the project meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria. The project will hence be recommended by SGS for registration with the UNFCCC.

Following the request for registration, the CDM EB requested a review of the project activity. The review process is described in the F-CDM-REG submitted with this request for registration. The issues raised in the request for review were all addressed to the satisfaction of the CDB EB Review Team.

Report No.: CDM.Val0018	Subject Group:	
Report title: GUJARAT FLUOROchemicals LTD VALIDATION OF CDM PROJECT FOR GHG EMISSION REDUCTION BY THERMAL OXIDATION OF HFC23		
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Indexing terms

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Abbreviations

AM – CDM Approved Methodology
CDM – Clean Development Mechanism
CER - Certified Emission Reductions
EIA – Environmental Impact Assessment
GFL – Gujarat Fluorochemicals Ltd
GHG – Greenhouse gas(es)
LPG – Liquefied Petroleum Gas
PDD - Project Design Document
UNFCCC – United Nations Framework Convention on Climate Change

Conversion Factors and Definitions

HFC23 – hydrofluorocarbon generated as a by-product in the manufacture of HCFC22.

HCFC22 – hydrofluorocarbon manufactured for use as a refrigerant gas.

Cut-off ratio – the limiting ratio of HFC23:HCFC22 designed to ensure that no unfair claim of credit can be made by the destruction of HFC23 in excess of that permitted under this CDM project.

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1 INTRODUCTION

This validation concerns a project to be implemented by Gujarat Fluorochemicals Ltd (GFL) in India to eliminate or substantially reduce emissions of the potent greenhouse gas HFC23. The objectives of the validation exercise are to confirm that the project meets the necessary CDM criteria, that the project follows the approved methodology (AM0001) and that the proposals presented by GFL in the Project Design Document (PDD) will lead to a realistic determination of the quantities of HFC23 destroyed. The scope of the validation covers the baseline methodology involving a review of historical data (required to assess the performance of the plant before implementation of the project and hence to determine cut-off or limiting ratio of HFC23 generation: HCFC22 production), and a study of the monitoring methodology and plan proposed for the assessment of the quantities of HFC23 destroyed and the emissions of greenhouse gases associated with the destruction process.

The project involves the installation of a thermal oxidation plant at the GFL site. This plant will destroy the HFC23 which would otherwise continue to be vented to atmosphere. The process will result in the emission of CO₂, both directly as a result of the destruction process and indirectly due to the use of LPG, steam and other inputs to the activity.

The validation was undertaken by SGS using an Assessment Team of three people, two of whom visited the GFL site in India. John Miles acted as Team Leader and concentrated on the measurement and data processing issues, whilst Shivananda Shetty reviewed the local legislative and other requirements as well as studying the sampling and laboratory procedures. Marco van der Linden did not visit the GFL site but undertook the review of the project against CDM and other relevant criteria. The recommendation was then subjected to a Technical Review by Gareth Phillips.

1.1 Objective

SGS has been commissioned to validate the CDM Project for GHG Emission Reduction by Thermal Oxidation of HFC23 at the HCFC Plant of Gujarat Fluorochemicals Ltd (GFL). The purpose of validation is to have an independent third party assess the project design. In particular, the Baseline Methodology, the Monitoring and Verification Protocol (MVP), and their compliance with relevant UNFCCC, host country and CDM criteria are validated in order to confirm that the project design as documented is sound and reasonable and meets the identified criteria. Validation is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of certified emission reductions (CERs).

UNFCCC criteria refer to the Kyoto Protocol criteria and the CDM rules and modalities as agreed in the Bonn Agreement and the Marrakech Accords.

1.2 Scope

The scope of the validation is to assess all aspects of GHG reduction involved in the project, including the project design, the baseline, the determination of the cut-off ratio and the procedures proposed for monitoring the emission reductions in the future.

The following documents were reviewed as part of the scope of the activity:
Project Design Document (PDD), including Baseline Study and Monitoring Plan
Approved Methodology (AM0001)
CDM procedures under the Kyoto Protocol and Marrakesh Accords

The validation is executed as a third party activity, aiming to give an objective and independent evaluation of the GFL project against CDM requirements, UNFCCC rules and associated interpretations. Consultancy for the project was provided by Ineos Fluor, the technology sponsors, and Price Waterhouse Coopers (PWC), the advisors to GFL for the project, who were engaged by GFL to write the PDD, including the Baseline Methodology and the Monitoring and Verification Protocol. PWC also advised on the project's compliance with relevant UNFCCC, host country and CDM criteria. Both Ineos Fluor and PWC were represented during the validation meetings.

The validation does not provide consultancy for the Client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design.

There is no official requirement for an Environmental Impact Assessment, however GFL have undertaken one voluntarily. The environmental impacts and Host Party requirements have been confirmed by the local assessor.

The validation process also reviews comments by local stakeholders and any responses given by GFL. Comments were invited from local communities, shareholders, employees and the labour unions. The local community representatives were concerned about the environmental impact of the project, GFL's role in conserving water resources and the employment potential of the project. The shareholders questioned the capital investment and its impact on GFL's bottom line; potential alternative uses for HFC23; possible liabilities for past venting of HFC23; and the requirements for new employees or re-training existing employees. The employees asked if the project would cause any deterioration in the working environment, whilst the labour unions stated that the company should not compromise the principles of sustainable development, enquired about the employment potential, asked about restrictions on emissions from the incinerator and enquired about occupational health and safety impacts of the project.

GFL have responded to all these comments and their responses are given in section G3 and annex 13 of the PDD.

Request for Review

Three members of the CDM EB raised questions regarding the request for registration. These were considered by the Executive Board at its Seventeenth Meeting and the Executive Board agreed to carry out a review of the project activity based on the scope which included the following three issues:

- a) Clarification whether the host country approval dated 9th November, 2004 replaced the host country approval dated 5th January 2004
- b) Disclaimer contained in the validation opinion issued by SGS United Kingdom Ltd.'s
- c) The possible implications of HCFC 22 as a greenhouse gas

These were addressed in the following way:

- a) The Host Party issued a letter dated 30th December, 2004 confirming that the letter issued on 9th November 2004 replaced the earlier letter of approval dated 5th January 2004. The letter dated 9th November 2004 contained all the elements required of a Host Party letter.
- b) SGS United Kingdom Ltd. agreed to remove the disclaimer on the grounds that all of the data used to determine the baseline emissions had been verified during the site visit, with the exception of two numbers, that relate to the GWP of HFC 23 and its molecular weight, both of which are established by scientific study and widely reported.
- c) As regards the possible implications of HCFC22 being a greenhouse gas:
 - a. The PPs corrected the statement in the CDM-PDD which said that “HCFC22 is not a greenhouse gas” to “Though HCFC 22 is not listed in the Annex A of the Kyoto Protocol as a greenhouse gas, it is recognized as a greenhouse gas by the IPCC”
 - b. The PPs explained that the emissions of HCFC22 are not regulated by the Host Party. This fact was also verified by the validation team to be in line with the Montreal Protocol.
 - c. The PPs submitted the Strategic Business Plan (prepared prior to knowledge of CDM) upto 2010 which showed that the production of HCFC22 was planned to increase within the existing production capacity and in line with historical growth trends. This fact was also confirmed by the validation team.
 - d. The PPs confirm that they would not claim any CERs relating to HCFC22 production in excess of its Strategic Business Plan, or actual HCFC22 production, whichever is lower
 - e. The PPs submitted detailed information on Environment Impact of the proposed project activity

1.3 GHG Project Description

The project involves the installation of a thermal oxidation unit at the HCFC22 plant operated by GFL at their site in Gujarat, India. This unit will destroy the HFC23 which would otherwise be vented to atmosphere, using thermal oxidation technology similar to that employed by Ineos Fluor, the technology sponsor of the project. It is anticipated that the new plant will be installed during 2004 and will be operating from the beginning of 2005.

The process will result in the emission of CO₂, both directly as a result of the destruction process and indirectly due to the use of LPG, steam, hydrated lime and caustic soda in the activity.

The project is estimated to produce a net reduction in GHG emissions of about 3,000,000 tonnes CO₂e per annum based on an estimated production of HCFC22 of 10,000 tonnes per annum.

2 METHODOLOGY

The validation was performed in the manner of an audit, where a desk review of the PDD was first undertaken against the Approved Methodology and CDM and other relevant criteria. The

desk review was followed by a site visit to GFL Gujarat. Some information was also obtained from selected experts in the field.

A risk based audit approach has been adopted where issues of critical importance to the successful validation have been addressed in more detail. This approach has led to the development of a validation protocol (VP) which serves the following purposes:

- It organises, details and clarifies the requirements the project is expected to meet; and
- It ensures a transparent validation process where the validator will document how a particular requirement has been validated and the result of the validation.

The validation protocol consists of three tables. The different columns in these tables are described in figure 1 below.

The completed validation protocol is enclosed in Appendix A to this report.

Validation Protocol Table 1: Mandatory Requirements			
Requirement	Reference	Conclusion	Cross reference
<i>The requirements the project must meet.</i>	<i>Gives reference to the legislation or agreement where the requirement is found.</i>	<i>This is either acceptable based on evidence provided (OK), or a Corrective Action Request (CAR) of risk or non-compliance with stated requirements. The corrective action requests are numbered and presented to the client in the Validation report.</i>	<i>Used to refer to the relevant checklist questions in Table 2 to show how the specific requirement is validated. This is to ensure a transparent Validation process.</i>

Validation Protocol Table 2: Requirement checklist				
Checklist Question	Reference	Means of verification (MoV)	Comment	Draft and/or Final Conclusion
<i>The various requirements in Table 1 are linked to checklist questions the project should meet. The checklist is organised in seven different sections. Each section is then further sub-divided. The lowest level constitutes a checklist question.</i>	<i>Gives reference to documents where the answer to the checklist question or item is found.</i>	<i>Explains how conformance with the checklist question is investigated. Examples of means of verification are document review (DR) or interview (I). N/A means not applicable.</i>	<i>The section is used to elaborate and discuss the checklist question and/or the conformance to the question. It is further used to explain the conclusions reached.</i>	<i>This is either acceptable based on evidence provided (OK), or a Corrective Action Request (CAR) due to non-compliance with the checklist question (See below). Clarification is used when the validation team has identified a need for further clarification.</i>

Table 3 refers to questions raised with the local assessor and the responses received.

Validation Protocol Table 4: Resolution of Corrective Action and Clarification Requests			
Draft report clarifications and corrective action requests	Ref. to checklist question in table 2	Summary of project owner response	Validation conclusion
<i>If the conclusions from the draft Validation are either a Corrective Action Request or a Clarification Request, these should be listed in this section.</i>	<i>Reference to the checklist question number in Table 2 where the Corrective Action Request or Clarification Request is explained.</i>	<i>The responses given by the Client or other project participants during the communications with the validation team should be summarised in this section.</i>	<i>This section should summarise the validation team's responses and final conclusions. The conclusions should also be included in Table 2, under "Final Conclusion".</i>

Figure 1 Validation protocol tables

2.1 Review of Documents

The Project Design Document submitted by the Client was reviewed against the Approved Methodology and against CDM and other relevant criteria. Additional background documents related to the project design and baseline were also made available during the site visit to GFL. These documents were also reviewed.

2.2 Follow-up Interviews

The site visit to the GFL site in Gujarat took place during December 2003. It was performed by the validators John Miles and Shivananda Shetty of SGS. During the site visit, issues relating to the HFC23 Thermal Oxidation Project were discussed with the following people:

- GFL
 - Deepak Asher – Vice President (Corporate Finance)
 - V K Soni – Head of Projects
 - Joseph Titus – Chief General Manager (Production)
 - D K Sachdeva – Vice President (Operations)
 - P S Parameswaran – General Manager (Technical)
 - Rajeev Gupta – Dy General Manager (Projects)
 - Akshay Parmar – General Manager (Engineering)
- Ineos Fluor / ICI
 - Pete Sleight – European Regulatory Manager – Ineos Fluor UK
 - Murali Duvvuri – Market Manager – ICI India
- PWC
 - Sameer K Singh
- Others
 - Lex de Jong, VROM, Govt of Netherlands (by telephone)

Technical details relating to the project were discussed with the appropriate GFL and Ineos Fluor personnel listed above. Operations staff were also interviewed to gain a better insight into the day to day operation of the existing GFL facilities.

Commercial and other wider issues were discussed with Deepak Asher, GFL Vice President. Compliance with environmental legislation in India was validated by document review and contact with relevant authorities and bodies in India, including the Ministry of Environment of Forests, the Gujarat Pollution Control Board, the regulatory handbook and discussion with the regulatory authorities.

2.3 Resolution of Clarification and Corrective Action Requests

During the meetings at GFL extensive discussions took place around the contents of the PDD. Where appropriate, clarification was sought from GFL and if any aspect of the report was considered to be unsatisfactory, or in need of reinforcement, GFL were asked to review it.



These items are reported under Clarifications and Corrective Action Requests in Table 3 of the Validation Protocol (Appendix A). All were resolved satisfactorily during or soon after the visit to GFL.

The direct findings of the validation are reported in the Validation Protocol and the conclusions are given in section 3 of this report. Where appropriate, further information concerning a request for clarification or corrective action is given in the Validation Protocol.

Since modifications to the Project design were necessary to resolve the questions raised by SGS, the Client decided to revise the documentation and re-issued the PDD on 8th March 2004. A final version of the PDD to include all project participants was issued on 23rd June, 2004. After reviewing the revised project documentation, SGS issued this final validation report and opinion.

3 VALIDATION FINDINGS

The key findings of the validation are summarised in the following sections. They are presented for each validation subject as follows:

- 1) The findings from the desk review of the original project design documents and the findings from interviews during the follow up visit are summarised. A more detailed record of these findings can be found in the Validation Protocol in Appendix A.
- 2) Where SGS identified issues that needed clarification, a Clarification Request was issued. The Clarification Requests are stated, where applicable, in the following sections and are further documented in the Validation Protocol in Appendix A. The validation of the Project resulted in ten Clarification Requests.
- 3) Where Clarification Requests have been issued, the exchanges between the Client and SGS to resolve these Clarification Requests are summarised.
- 4) The conclusions for each validation subject are presented.

The final validation findings relate to the project design as documented and described in the revised and resubmitted project design documentation.

3.1 Project Design

The technology employed in the project is similar to that deployed by Ineos Fluor in their plant at Runcorn UK. This technology also forms the basis of the Approved Methodology (AM0001), which was developed for a similar project in Ulsan, Korea. The technology employed by Ineos Fluor is regarded as state of the art and its application in Runcorn is subject to stringent inspection for compliance with UK environmental regulations, and separately under the UK Emissions Trading Scheme. It is therefore reasonable to argue that the project employs current best practice and that, on the experience gained in Runcorn, the project will produce the maximum achievable reduction in GHG emissions.

Both project duration and crediting time are clearly described in the PDD and are considered reasonable for this type of technology.

The project's contribution to sustainable development is achieved by increasing local incomes in the region of Gujarat in which the project will be based and by reducing vulnerability whilst increasing empowerment of vulnerable sections of society. In parallel GFL are committed to the improvement of natural resources in the vicinity of the project activity.

3.2 Baseline

The PDD describes the Baseline Methodology which is in conformance with the Approved Methodology (AM 0001). The key conclusions are summarised below, but further details are given in the Validation Protocol.

There are no regulations in India requiring the destruction of HFC23 and there is no market for the sale of HFC23 in India. It is therefore realistic to argue that the baseline condition should be that where all the HFC23 generated during HCFC22 production is being emitted to atmosphere.

It is accepted that the chloroform mass balance should be used to derive the cut-off ratio, which should be set at 2.9%, the lowest ratio achieved over the last three years. The revised PDD provides the justification for this approach.

It is not mandatory for GFL to install the thermal decomposition facility and, if installed, the facility will involve economic/financial commitment for GFL, consisting of capital and operating (recurring) costs without economic benefits besides the CER's. Therefore it is justified that the project activity itself is not a likely baseline scenario.

All the relevant GHG sources are identified, the system boundaries defined and the methods of determining the emissions are adequately described. Using these methods, estimates of emissions reductions using an illustrative production of HCFC22 of 10000MT per year show that emissions reductions due to the project will be in excess of 3 million tonnes CO₂e per year. The emission reductions will increase in future years in tandem with HCFC22 production in line with the company's business plan.

3.3 Monitoring Plan

It is considered that the Monitoring Plan will provide the necessary information to monitor and report emissions reductions due to the project. The protocol will allow the contribution from all sources to be identified in relation to the project operation and baseline factors.

The frequency of recording for parameters to be measured or calculated in connection with the project activity is specified in the PDD and ranges from monthly to yearly in line with the requirements of the Approved Methodology. Certain calibration checks and other tests on the installed instrumentation will be carried out more frequently than specified in the Approved Methodology, thereby providing improved accuracy and confidence in the measurements.

The responsibility within GFL for monitoring and reporting data relating to all project emissions and emissions reductions is defined in the PDD. The measurement and monitoring procedures in use at the existing GFL plant in Gujarat will be extended to cover the new thermal oxidation unit and the QA manual will be modified accordingly.

Monitoring of Environmental Impacts is not explicit in the monitoring plan but is subject to environmental legislation and control by the Gujarat Pollution Control Board.

The PDD discusses the project's contribution to sustainable development. Whilst the monitoring plan does not currently identify indicators of sustainable development nor provide for monitoring of this contribution, GFL has agreed to identify such indicators and means of monitoring them, before the first verification. An Observation was raised to highlight this issue to the project developer and the verifier.

3.4 Calculation of GHG Emissions

The calculation methods applied to the determination of GHG emissions are explained in detail in the PDD. They follow the procedures laid down in the Approved Methodology.

All the relevant GHG sources are identified and accounted for in the PDD and the project boundaries are properly defined. The formulae and factors used for the determination of GHG emissions follow accepted practice and, where possible, conversion and other factors have been compared with those used in similar projects elsewhere. No significant differences were found.

The calculation of emissions reductions is based primarily on the measurement of HFC23 generated during the production of HCFC22. This measurement is based on the use of state of

the art technology, ensuring that the measurement uncertainty will be as low as is practicably possible.

It was noted that although an allowance has been made for the emissions generated by the transport of sludge to landfill, no similar allowance had been made in the PDD or AM for the transport of LPG to the GFL site. GFL argued that these emissions could be ignored for the following reasons:

- The emissions will be smaller than those associated with sludge transport
- The emissions are only a very small part of the those generated by the project, which are in turn approx 0.04% of the reduction in emissions.
- The emissions will be outside the control of GFL and will be difficult to determine.

This argument is realistic and is therefore accepted.

3.5 Environmental Impacts

The PDD states that an Environmental Impact Assessment (EIA) for the proposed project has been undertaken. Existence of the report has been verified during the site visit and a copy made available for inspection subsequently.

Under Indian Law when a project is below Rs 50 crores and is not in the red category industry, approval of the EIA by the local pollution board is required. The project at GFL falls within this limit. A "Certificate of No Objection" by the Gujarat Pollution Control Board was provided dated 15 March 2004 (see Appendix C).

3.6 Comments by Local Stakeholders

GFL invited comments from local communities, shareholders, employees and the labour unions at meetings held at their Vadodara office on 13th June 2003. The responses from GFL are summarised below.

The local community representatives' concerns about the environmental impact of the project, GFL's role in conserving water resources and the employment potential of the project, were answered as follows:

- The project will conform to the best environmental performance standards. An EIA will also be conducted and a plan drawn up to ensure minimal adverse environmental impact.
- Most of the water required by the proposed system will be recycled, although about 15 m³/day will be drawn from external sources. GFL will continue their initiatives to improve water management in the region.
- Between 30 and 40 new jobs will be created by the Project and where possible these will be offered to local people.

The shareholders' questions concerning the capital investment and its impact on GFL's bottom line; potential alternative uses for HFC23; possible liabilities for past venting of HFC23; and the requirements for new employees or re-training existing employees were answered as follows:

- The project may have a positive impact on GFL's bottom line from the revenue received through the transfer of CERs.
- The alternative uses for HFC23 are limited and there are no known uses in India.

- There are no regulations in India restricting emissions of HFC23 since it is not toxic. Hence past emissions are unlikely to create any liabilities.
- Between 30 and 40 new employees will be recruited.

The questions raised by the employees and the labour unions were answered as follows:

- In response to employee concerns about degradation of the working environment, GFL advised that the technology supplier had assured them that there will be no leakage of HF and HCl. However, GFL will monitor the workplace for these materials.
- In discussion with the labour union representative, GFL advised that the Project will conform to relevant emission standards prescribed in EU directives; that 30-40 new employment opportunities will be created, about 90% of which will be unskilled; and the possibility of escape of toxic gases to atmosphere is unlikely since the process is a closed system.

4 COMMENTS BY PARTIES, STAKEHOLDERS AND NGOS

SGS published the project documents on its website from 2003/12/01 until 2003/12/31 and invited comments by Parties, stakeholders and non-governmental organisations. No relevant comments were received during the 30 day period.

5 VALIDATION OPINION

SGS has performed a validation of the CDM project for GHG emissions reduction by thermal oxidation of HFC23 at Gujarat Fluorochemicals Ltd, India. The validation was performed on the basis of UNFCCC criteria and host country criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The review of the project design documentation and the subsequent follow-up interviews have provided SGS with sufficient evidence to determine the fulfilment of the stated criteria. In our opinion, the project meets all relevant UNFCCC requirements for the CDM and all relevant host country criteria. The project will hence be recommended by SGS for registration with the UNFCCC.

SGS has received confirmation by the host Party that the project activity assists it in achieving sustainable development.

By destroying HFC23 (a potent greenhouse gas) generated during the production of HCFC22, the project results in reductions of greenhouse gas emissions that are real, measurable and give long-term benefits to the mitigation of climate change. An analysis of the investment and technological barriers demonstrates that the proposed project activity is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity. Given that the project is implemented as designed, the project is likely to achieve the estimated amount of emission reductions.

6 REFERENCES

Category 1 Documents:

- 1 CDM Project Design Document "PROJECT FOR GHG EMISSION REDUCTION BY THERMAL OXIDATION OF HFC 23 at HCFC22 PLANT of GUJARAT FLUOROCHEMICALS LIMITED (GFL)" prepared by Price Waterhouse Coopers, dated November 14 2003, updated 8 March 2004 and 23 June 2004.
- 2 Host Country Letter of Approval dated 5 January 2004.
- 3 Gujarat Pollution Control Board "No Objection Certificate" dated 15 March 2004.
- 4 EIA Report (confidential)
- 6 GFL Strategic Business Plan (confidential)
- 7 Ozone Depleting Substances (Regulation and Control) Rules, 2000, Government of India

Category 2 Documents:

- 8 CDM Approved Methodology AM0001, Incineration of HFC23 Waste streams, dated 26 September 2003.
- 9 Government of India – The Environment (Protection) Act 1986

Persons interviewed:

All those interviewed are listed in section 2.2 of this report.

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